Lecture 3:

Maximum Likelihood Estimation (MLE), Linear Regression, Linear Models

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Bayesian Data Analysis and Machine Learning for Physical Sciences

Course Map

Module 1	Maximum Entropy and Information, Bayes Theorem
Module 2	Naive Bayes, Bayesian Parameter Estimation, MAP
Module 3	**MLE, Lin Regression, Model selection: Comparing Distributions**
Module 4	Model Selection: Bayesian Signal Detection
Module 5	Variational Bayes, Expectation Maximization
Module 6	Stochastic Processes
Module 7	Monte Carlo Methods
Module 8	Markov Models, Graphs
Module 9	Machine Learning Overview, Supervised Methods
Module 10	Unsupervised Methods
Module 11	ANN: Perceptron, Backpropagation
Module 12	ANN: Basic Architecture, Regression vs Classification, Backpropagation again
Module 13	Convolution and Image Classification and Segmentation
Module 14	TBD (GNNs)
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Module 16	TBD (Transformer and LLMs)

Bayesian Model Selection/Testing

Outline

Standard Tests

- the p-value

Bayesian Model Testing

- idea
- curve fitting revisited
- comparing distributions

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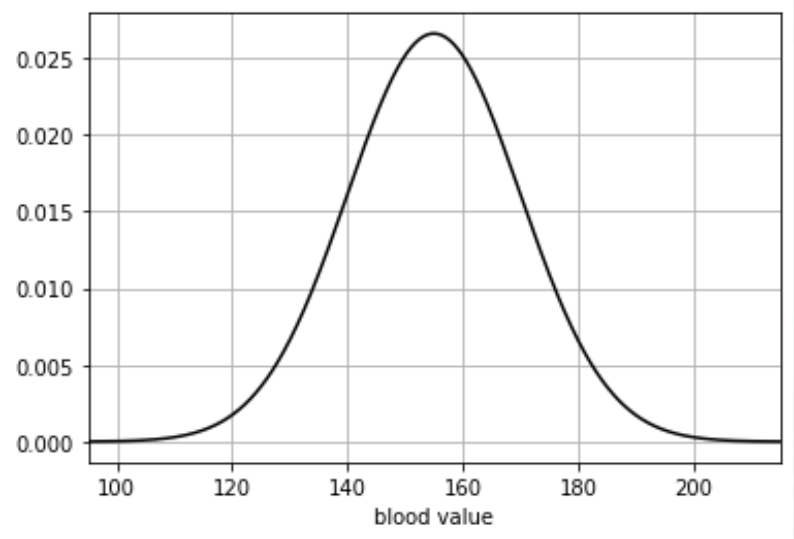
Bayesian Model Selection/Testing: the p-value

standard frequentist way (not Bayesian):

1) assume a **likelihood function *L* as your model**, aka **null hypothesis H_0**
2) take a datapoint x_0
3) calculate the ***probability P*** given ***L*** i. e. **given H_0** is true, **that x_0 has this value** or a **more extreme value**
4) accepting or rejecting H_0 based on ***P*** and the threshold α

example: 1) a healthy person has a blood value that follows a **normal distribution** with $\mu = 155, \sigma = 15$, i. e. $\boldsymbol{H_0}: N(\boldsymbol{\mu = 155, \sigma = 15})$



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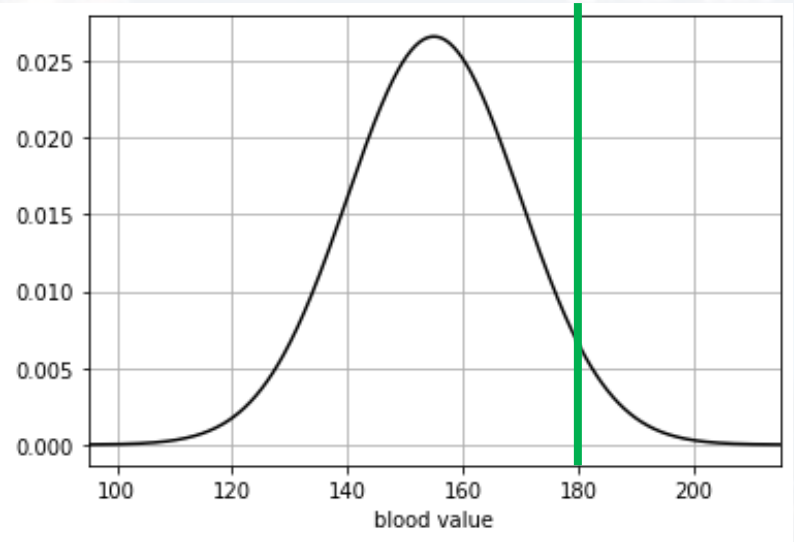
2) a patient has the value $\boldsymbol{x_0 = 180}$



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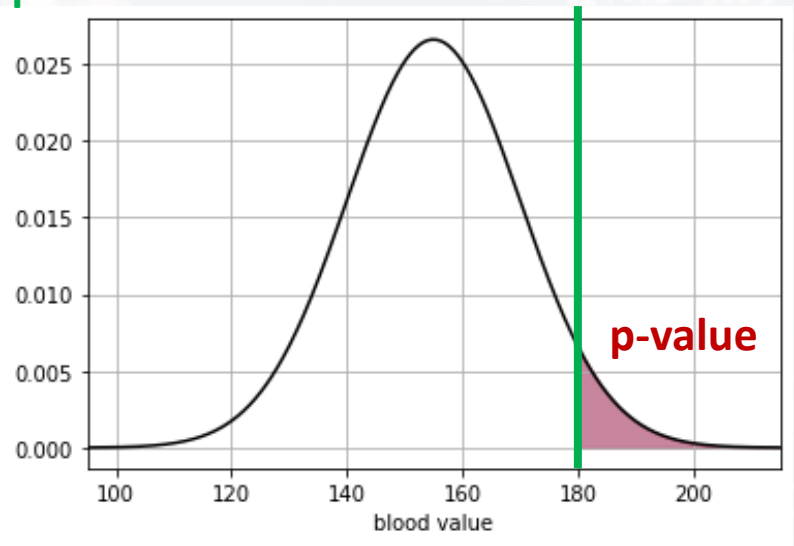
2) a patient has the value x_0 **= 180**

3) probability $\boldsymbol{P(x \geq 180 | H_0) = 0.048}$ called **p-value**



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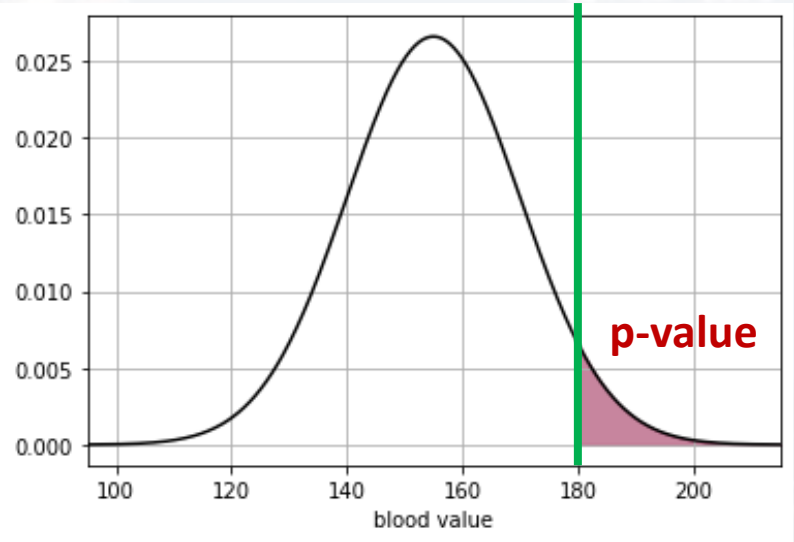
4) for **α = 0.05**, H_0 is **rejected**, i. e. patient is **not healthy**

→ alternative hypothesis $\boldsymbol{H_1}$: **not healthy**



Bayesian Model Selection/Testing:

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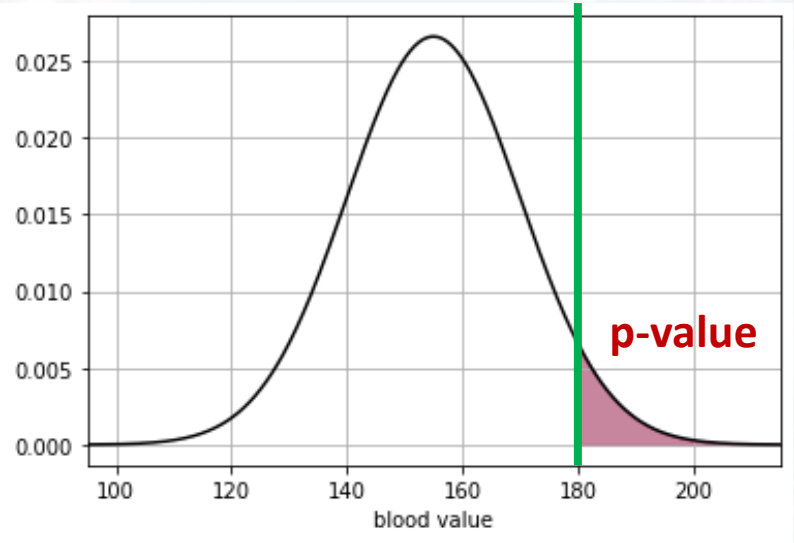
We just performed a so-called Z – Test
(comparing one value to a normal distribution)



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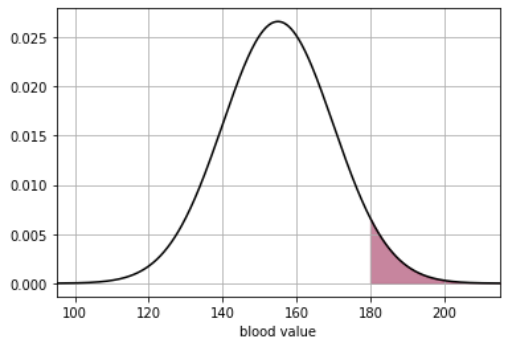
caveats:

- the model ***L*** for H_0 has to **be known**
- the p-value $P(x \geq x_0 | H_0)$ **does not** tell if H_0 is true or not
- the p-value $P(x \geq x_0 | H_0)$ **does not** tell which hypothesis is more likely
- the p-value just gives $P(x \geq x_0 | H_0)$
- the threshold α for accepting/rejecting H_0 is **arbitrary**
- we are aiming on disproving a hypothesis, by assuming it is true, **without** leading to a contradiction

Bayesian Model Selection/Testing:

- data point *x* versus **normal**
- $H_0: x \in N(\mu, \sigma)$

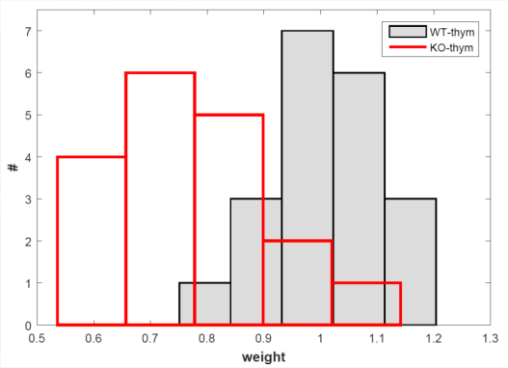
Z-test



- **normal** versus another **normal**
- two samples of sizes n_1 and n_2

t-test

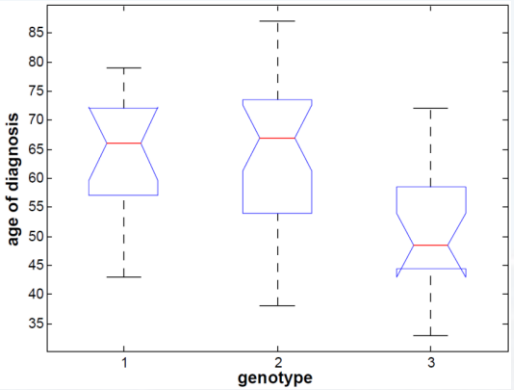
- two tail $H_0: \mu_1 = \mu_2, H_1: \mu_1 \neq \mu_2$
- right tail $H_0: \mu_1 < \mu_2, H_1: \mu_1 > \mu_2$
- left tail $H_0: \mu_1 > \mu_2, H_1: \mu_1 < \mu_2$



- **N normal dist.**
- N samples of sizes n_i

ANalysis **O**f **VA**riance

- $H_0: \mu_i = \mu_j \; \forall \, i, j$, $H_1: \mu_i \neq \mu_j$ for at least one pair i, j

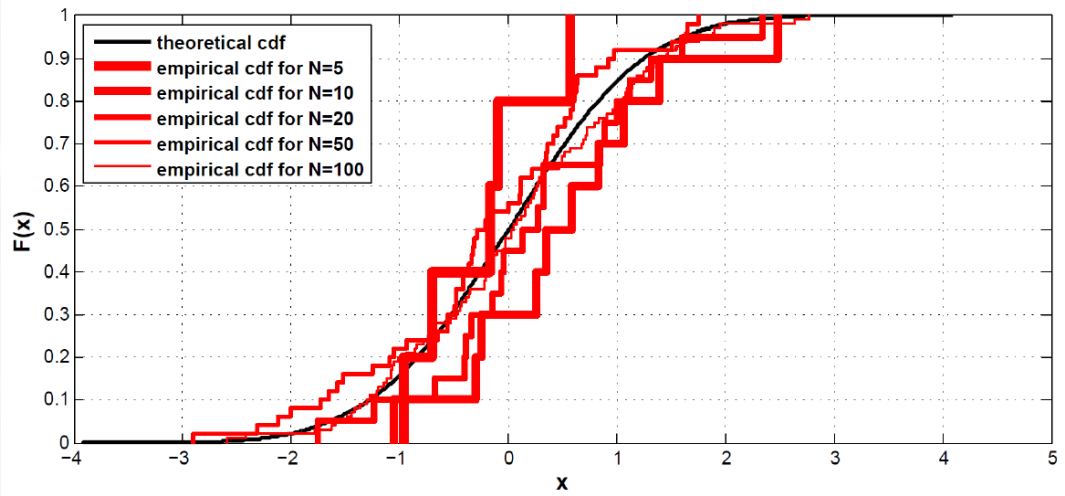


Bayesian Model Selection/Testing:

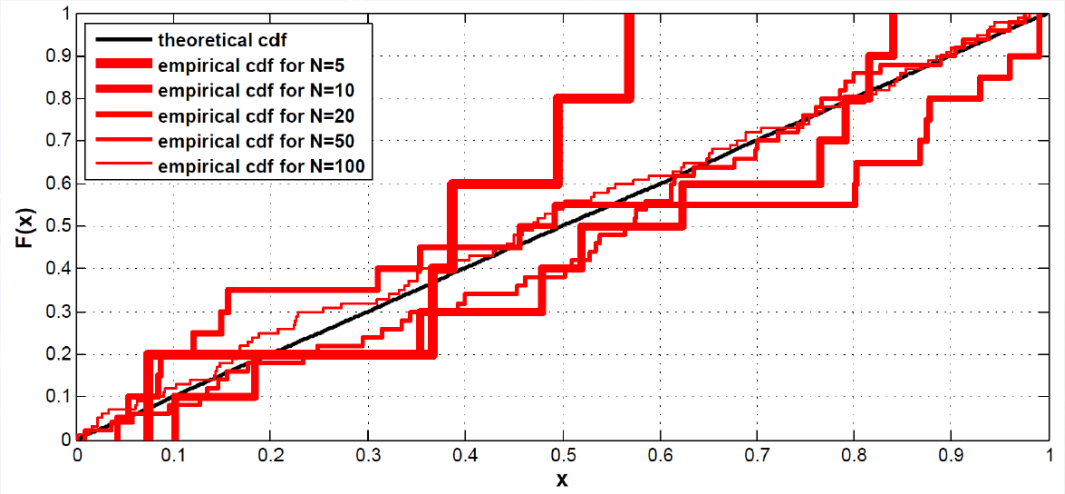
testing if a data set follows a particular distribution

Kolmogorov – **S**mirnov – test (KS test)

example: normal distribution:



example: uniform distribution:



Bayesian Model Selection/Testing:

the p-value

testing if a data set follows a particular distribution

ranking tests (Wilcoxon)

…and many more…

They all generate a test statistic: Z-value, t-value, F-value etc

→ from that: calculating a p-value

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Bayesian Mod

the p-value

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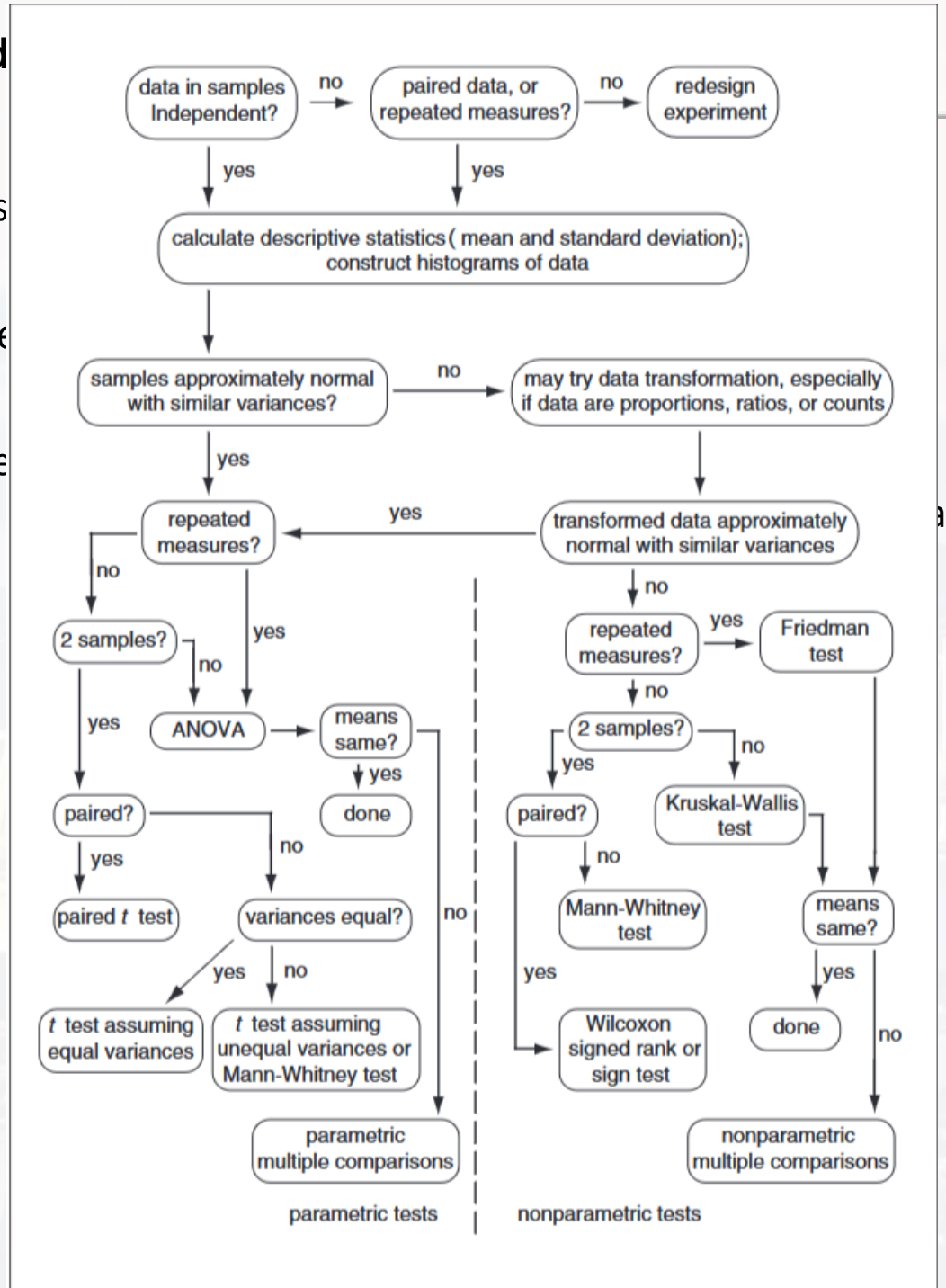
alue



Current Protocols in Molecular Biology (1998) A.3I.1-A.3I.22

Bayesian Model Selection/Testing

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- idea
- curve fitting revisited
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Bayesian Model Selection/Testing:

often, we have many competing models $\mathrm{M_i}$

goal: $$\rho = \frac{P(\mathrm{M_A}|D,I)}{P(\mathrm{M_B}|D,I)}$$ odds ratio

$$= \frac{\boxed{P(D|M_A,I)}\,P(M_A|I)}{P(D|I)} \cdot \frac{P(D|I)}{\boxed{P(D|M_B,I)}\,P(M_B|I)}$$ Bayes' theorem

idea
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D:	data set
I:	information
M_A:	model A
M_B:	model B

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$$= \frac{\boxed{P(D|M_A,I)}\,P(M_A|I)}{P(D|I)} \cdot \frac{P(D|I)}{\boxed{P(D|M_B,I)}\,P(M_B|I)}$$

D:	data set
I:	information
M_A:	model A
M_B:	model B
$\{\alpha\}_i$:	all parameter of model M_i

marginalization:

$$P(D|M_i,I) = \int P(D,\{\alpha\}_i|M_i,I)\ d\Omega_{\{\alpha\}_i}$$

$$P(x,y) = P(x|y)P(y)$$

$$= \int P(D|\{\alpha\}_i,M_i,I)\ P(\{\alpha\}_i|M_i,I)\ d\Omega_{\{\alpha\}_i}$$

$$= \int P(D|\{\alpha\}_i,M_i,I) \prod_j P(\alpha_{ij}|M_i,I)\ d\alpha_{ij}$$

assuming all α_{ij} are mutually independent **(Naïve Bayes)**

Bayesian Model Selection/Testing:

$$\boldsymbol{P(D|M_i,I)} = \int \underbrace{P(D|\{\alpha\}_i, M_i, I)}_{\text{likelihood function}} \prod_j \underbrace{P(\alpha_{ij}|M_i, I)}_{\text{prior of } \alpha_{ij}} \, d\alpha_{ij}$$

likelihood function
→ the actual model

prior of $\boldsymbol{\alpha_{ij}}$ BEVORE(!) measurement

maximum entropy without prior knowledge: $\frac{1}{\boldsymbol{\alpha_{ij}}(max) - \boldsymbol{\alpha_{ij}}(end)}$

idea
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$\boldsymbol{D}$:	**data set**
$\boldsymbol{I}$:	**information**
$\boldsymbol{M_A}$:	**model A**
$\boldsymbol{M_B}$:	**model B**
$\{\boldsymbol{\alpha}\}_i$:	**all parameter of model $\boldsymbol{M_i}$**

$$\boxed{\rho = \frac{P(\mathrm{M_A}|D,I)}{P(\mathrm{M_B}|D,I)} = \frac{P(M_A)}{P(M_B)} \cdot \frac{\int P(D|\{\alpha\}_A, M_A, I)\, d\Omega_{\{\alpha\}_A}}{\int P(D|\{\alpha\}_B, M_B, I)\, d\Omega_{\{\alpha\}_B}} \cdot \frac{\prod_j \boldsymbol{\alpha_{jB}}(max) - \boldsymbol{\alpha_{jB}}(min)}{\prod_j \boldsymbol{\alpha_{jA}}(max) - \boldsymbol{\alpha_{jA}}(min)}}$$

likelihood function
→ the actual model

Occam's Razor:
simple models are preferred

prior probability of each model: maximum entropy → 1:1

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$$\rho = \frac{P(\mathrm{M_A}|D,I)}{P(\mathrm{M_B}|D,I)} = \frac{P(M_A)}{P(M_B)} \cdot \frac{\int P(D|\{\alpha\}_A, M_A, I)\, d\Omega_{\{\alpha\}_A}}{\int P(D|\{\alpha\}_B, M_B, I)\, d\Omega_{\{\alpha\}_B}} \cdot \frac{\prod_j \boldsymbol{\alpha_{jB}}(max) - \boldsymbol{\alpha_{jB}}(min)}{\prod_j \boldsymbol{\alpha_{jA}}(max) - \boldsymbol{\alpha_{jA}}(min)}$$

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curve fitting revisited
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→ the actual model

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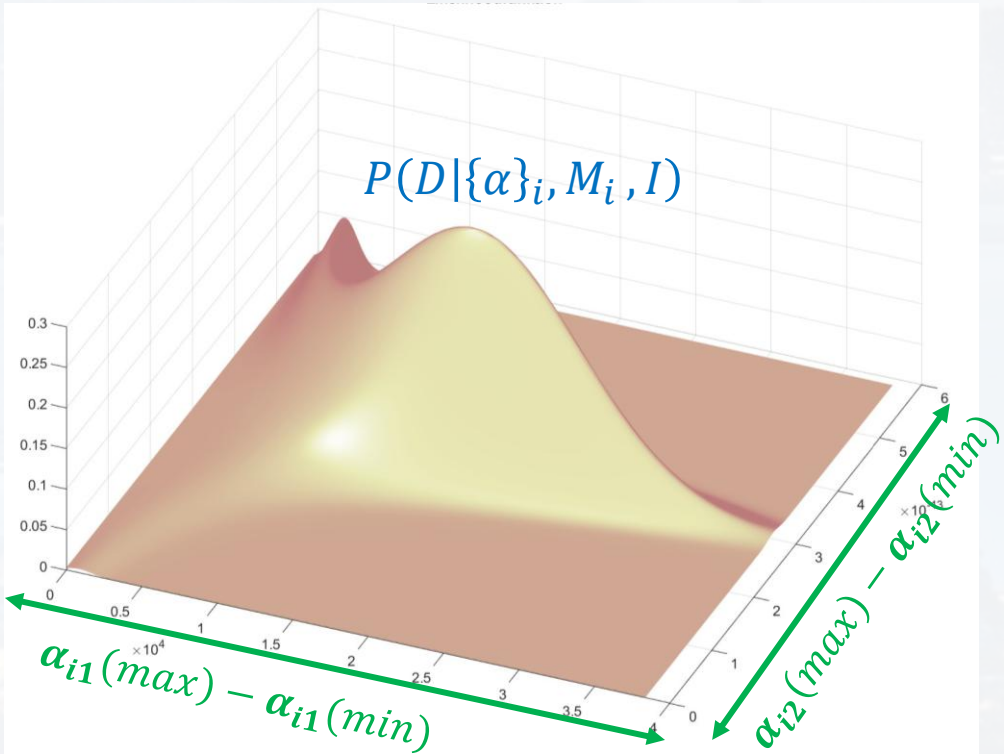
setting $\boldsymbol{\alpha_{ji}}(max) - \boldsymbol{\alpha_{ji}}(min)$ such that

- it covers the **entire integral** over $P(D|\{\alpha\}_i, M_i, I)$
- it makes **physical sense**
  (e. g. $\omega_{max} \approx$ Nyquist frequency,
  $\omega_{min} \approx 2/T_{obs}$)



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$$P(\alpha|D, M_i) = \frac{P(D|\alpha, M_i)P(\alpha|M_i)}{\boxed{P(D|M_i)}}$$

BPE: $P(\alpha|D) = \frac{P(D|\alpha)P(\alpha)}{P(D)}$

$P(D|M_i)$ is called **evidence** for the model M_i
(usually omitted for BPE)

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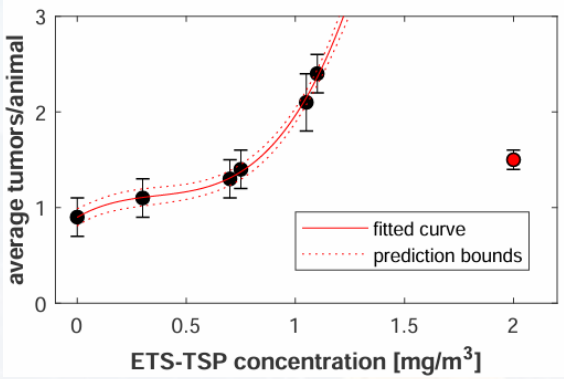
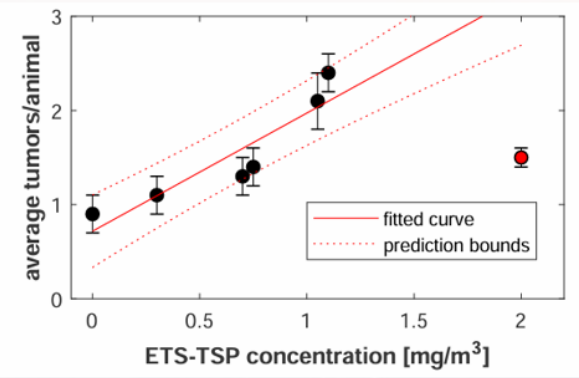
model $\mathrm{M_A}$



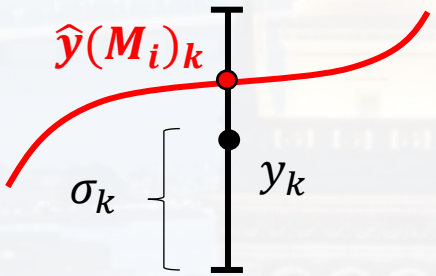
model $\mathrm{M_B}$



assumptions:

1) considerably good fit for each model M_i (otherwise, we wouldn't need Bayesian methods)

2) as before: y_k are drawn from a Gaussian $\mathcal{N}(y_k, \sigma_k)$



$$P(y_k|\{\alpha\}_i, M_i, I) \approx \frac{1}{\sqrt{2\pi\sigma_k^2}}\, e^{-\frac{1}{2}\frac{(y_k - \hat{y}(\boldsymbol{M_i})_k)^2}{\sigma_k^2}}$$

for $\sigma_k \ll |y_k|$

$\boldsymbol{y_k}$: **measured value**

$\boldsymbol{\sigma_k}$: **error**

$\hat{y}(\mathrm{M_i})_\mathrm{k}$: **model value (after fit)**

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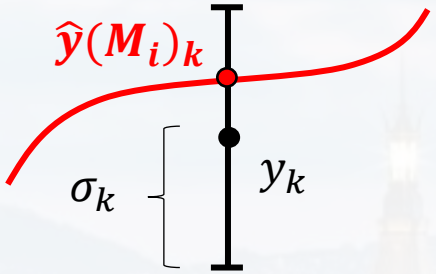

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$$P(D|\{\alpha\}_i, M_i, I) = \prod_k P(y_k|\{\alpha\}_i, M_i, I) = \prod_k \frac{1}{\sqrt{2\pi\sigma_k^2}}\, e^{-\frac{1}{2}\frac{(y_k - \hat{y}(M_i)_k)^2}{\sigma_k^2}}$$

$$= \left(\prod_k \frac{1}{\sqrt{2\pi\sigma_k^2}}\right) \cdot e^{-\frac{1}{2}\sum_k \frac{(y_k - \hat{y}(M_i)_k)^2}{\sigma_k^2}}$$

$$= \left(\prod_k \frac{1}{\sqrt{2\pi\sigma_k^2}}\right) \cdot e^{-\frac{1}{2}\chi_i^2}$$

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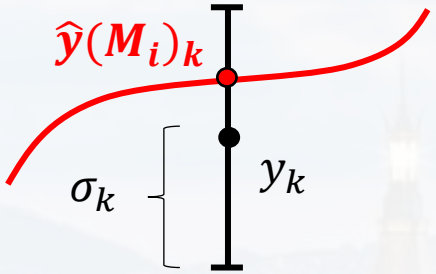

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$$P(D|\{\alpha\}_i, M_i, I) = \left(\prod_k \frac{1}{\sqrt{2\pi\sigma_k^2}}\right) \cdot e^{-\frac{1}{2}\chi_i^2}$$

$$\rho = \frac{P(\mathrm{M_A}|D,I)}{P(\mathrm{M_B}|D,I)} = \frac{P(M_A)}{P(M_B)} \cdot \frac{\int e^{-\frac{1}{2}\chi_A^2}\, d\Omega_{\{\alpha\}_A}}{\int e^{-\frac{1}{2}\chi_B^2}\, d\Omega_{\{\alpha\}_B}} \cdot \frac{\prod_j \boldsymbol{\alpha_{jB}}(max) - \boldsymbol{\alpha_{jB}}(min)}{\prod_j \boldsymbol{\alpha_{jA}}(max) - \boldsymbol{\alpha_{jA}}(min)}$$

Not only χ_i^2, but also model complexity (Occam's razor) and priors are important!

Bayesian Model Selection/Testing:

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note: - like for MLE we found χ_i^2, if y_k are drawn from $\mathcal{N}(y_k, \sigma_k)$ but this time weighted with priors

- $\int e^{-\frac{1}{2}\chi_i^2}\, d\Omega_{\{\alpha\}_i}$ is usually hard to evaluate

→ numerically, or Monte-Carlo (see later), or

- $\chi_i^2 = \chi_i^2(\{\alpha\}_i)$ reaches a minimum for a certain set of $\{\alpha\}_i$, $\{\bar{\alpha}\}_i$

- 2nd order Taylor approx.:

$$\chi_i^2(\{\alpha\}_i) \approx \chi_i^2(\{\bar{\alpha}\}_i) + 0 + \frac{1}{2}(V_i - \bar{V}_i)^T\, \nabla\nabla\chi_i^2(\{\bar{\alpha}\}_i)(V_i - \bar{V}_i)$$

where V_i and $\bar{V}_i$ are vectors containing $\{\alpha\}_i$ and $\{\bar{\alpha}\}_i$, respectively

Bayesian Model Selection/Testing

Outline

Standard Tests

- the p-value

Bayesian Model Testing

- idea
- curve fitting revisited
- comparing distributions

Bayesian Model Selection/Testing:

idea
curve fitting revisited
comparing distributions

problem: - often we want to know if two (or more) data sets D_m and D_n have been drawn from the same distribution (comparing μ or σ^2, like t-test or ANOVA)

- p – value **does not** answer the actual question!

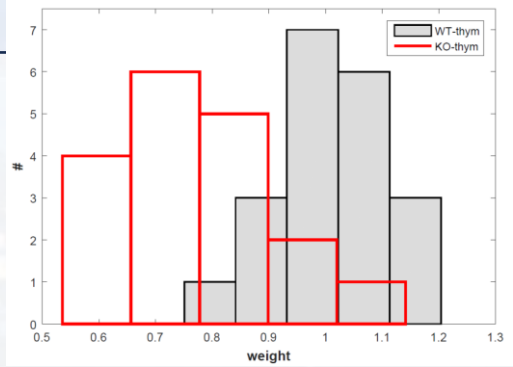
goal: - we want to be able to make the following statements:

- what is the probability that $\mu_m > \mu_n$
  $\sigma_m > \sigma_n$ … etc

- what is more likely: that D_m and D_n have been drawn from the same distribution or from two different distributions

idea: calculating the **evidence** via **marginalization!**

$$\rho = \frac{P(\mathrm{M_A}|D,I)}{P(\mathrm{M_B}|D,I)}$$

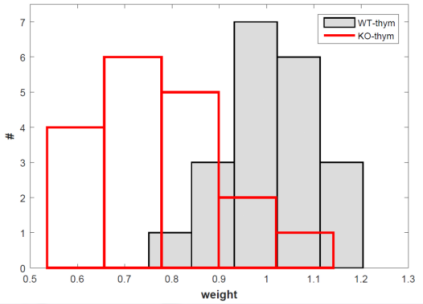


Bayesian Model Selection/Testing:

idea
curve fitting revisited
comparing distributions

example: $\mathcal{N}(y_k, \sigma_k)$

a) model A: data sets D_m and D_n have been drawn from the same distribution



$$P(D_m, D_n | A, I) = \iint P(D_m, D_n | \mu, \sigma, A, I)\, P(\mu, \sigma | A, I)\, d\mu d\sigma$$

if **max ent**:

$$P(\mu, \sigma | A, I) = \frac{1}{\mu_{max} - \mu_{min}} \frac{1}{\sigma_{max}}$$

likelihood function
→ the actual model

D_m: N_m data points
D_n: N_n data points

$$P(D_m, D_n | \mu, \sigma, A, I) = \prod_{i=1}^{N_{tot}} P_i(D_m, D_n | \mu, \sigma, A, I)$$

$$N_{tot} = N_n + N_m$$

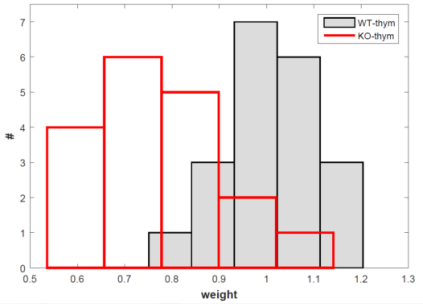
$$= \frac{1}{(2\pi\sigma^2)^{N_{tot}/2}} \cdot e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{N_{tot}}(x_i-\mu)^2}$$

Bayesian Model Selection/Testing:

idea
curve fitting revisited
comparing distributions

example: $\mathcal{N}(y_k, \sigma_k)$

a) model A: data sets D_m and D_n have been drawn from the same distribution



$$P(D_m, D_n | A, I) = \frac{1}{\mu_{max} - \mu_{min}} \frac{1}{\sigma_{max}} \iint \frac{1}{(2\pi\sigma^2)^{N_{tot}/2}} \cdot e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{N_{tot}}(x_i-\mu)^2} \, d\mu d\sigma$$

D_m:	N_m data points
D_n:	N_n data points

$N_{tot} = N_n + N_m$

again, the integral is usually hard to evaluate:

→ numerically, or

→ Taylor approximation for large N_{tot} around $\hat{\sigma}^2$ and $\hat{\mu}$

where

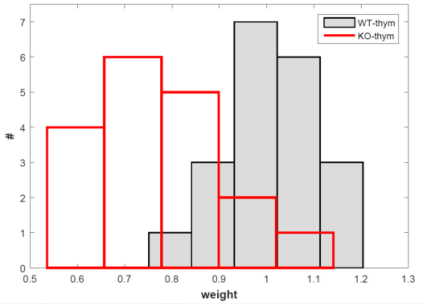
$$\hat{\sigma}^2 = \frac{1}{N_{tot}} \sum_{i=1}^{N_{tot}} (x_i - \hat{\mu})^2 \quad \text{and} \quad \hat{\mu} = \frac{1}{N_{tot}} \sum_{i=1}^{N_{tot}} x_i$$

Bayesian Model Selection/Testing:

idea
curve fitting revisited
comparing distributions

example: $\mathcal{N}(y_k, \sigma_k)$

a) model A: data sets D_m and D_n have been drawn from the same distribution



$$P(D_m, D_n | A, I) = \frac{1}{\mu_{max} - \mu_{min}} \frac{1}{\sigma_{max}} \iint \frac{1}{(2\pi\sigma^2)^{N_{tot}/2}} \cdot e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{N_{tot}}(x_i-\mu)^2} \, d\mu d\sigma$$

again, the integral is usually hard to evaluate:

→ numerically, or

→ Taylor approximation for large $\boldsymbol{N_{tot}}$ around $\hat{\sigma}^2$ and $\hat{\mu}$

$$P(D_m, D_n | A, I) \approx \frac{1}{\sqrt{2}} \frac{\left(\hat{\sigma}\sqrt{2\pi}\right)^{2-N_{tot}} e^{-N_{tot}/2}}{(\mu_{max} - \mu_{min})\, \sigma_{max}\, \boldsymbol{N_{tot}}}$$

$\boldsymbol{D_m}$: $\boldsymbol{N_m}$ **data points**
$\boldsymbol{D_n}$: $\boldsymbol{N_n}$ **data points**

$\boldsymbol{N_{tot} = N_n + N_m}$

$$\hat{\mu} = \frac{1}{N_{tot}} \sum_{i=1}^{N_{tot}} x_i$$

$$\hat{\sigma}^2 = \frac{1}{N_{tot}} \sum_{i=1}^{N_{tot}} (x_i - \hat{\mu})^2$$

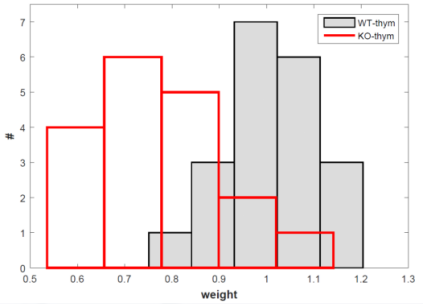
Bayesian Model Selection/Testing:

idea
curve fitting revisited
comparing distributions

example: $\mathcal{N}(y_k, \sigma_k)$

b) model B: data sets D_m and D_n have been drawn from **different** distributions

$$P(D_m, D_n | B, I) = P(D_m | B, I)\, P(D_n | B, I)$$



in general:

$$P(\{D_k\} | B, I) = \prod_{k=1}^{K} P(D_k | B, I)$$

D_m: N_m data points
D_n: N_n data points

$N_{tot} = N_n + N_m$

as before: marginalization over evidence term **(assuming same ranges for μ and σ)**

$$P(D_k | B, I) = \frac{1}{\mu_{max} - \mu_{min}} \frac{1}{\sigma_{max}} \iint \frac{1}{[2\pi\, \sigma^2(k)]^{N_k/2}} \cdot e^{-\frac{1}{2\sigma^2(k)} \sum_{i=1}^{N_k} (x_i - \mu(k))^2}\, d\mu(k)\, d\sigma(k)$$

now: comparing odds ratios!

Bayesian Model Selection/Testing:

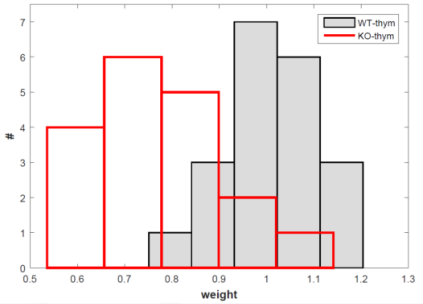
idea
curve fitting revisited
comparing distributions

example: $\mathcal{N}(y_k, \sigma_k)$

now: comparing odds ratios!

$$\rho = \frac{P(\mathrm{M_A}|D,I)}{P(\mathrm{M_B}|D,I)} = \frac{P(D_m, D_n|\, A, I)}{P(D_m, D_n|\, B, I)} \frac{P(A|I)}{P(B|I)}$$

$$= \frac{(\mu_{max} - \mu_{min})\, \sigma_{max}}{\sqrt{2\pi}} \cdot \frac{N_m\, N_n}{N_{tot}} \cdot \frac{(\hat{\sigma})^{2-N_{tot}}}{(\hat{\sigma}_n)^{2-N_n}\, (\hat{\sigma}_m)^{2-N_m}} \cdot \frac{P(A|I)}{P(B|I)}$$



D_m: N_m data points
D_n: N_n data points

$N_{tot} = N_n + N_m$

for *K* different distributions:

$$P(\{D_k\}|\, B, I) = \prod_{k=1}^{K} P(D_k|\, B, I)$$

$$\rho = \left[\frac{(\mu_{max} - \mu_{min})\, \sigma_{max}}{\sqrt{2\pi}}\right]^{K-1} \cdot \frac{(\hat{\sigma})^{2-N_{tot}}}{N_{tot}} \cdot \prod_{k=1}^{K} \frac{N_k}{(\hat{\sigma}_k)^{2-N_k}} \cdot \frac{P(A|I)}{P(B|I)}$$

Bayesian Model Selection/Testing:

idea
curve fitting revisited
comparing distributions

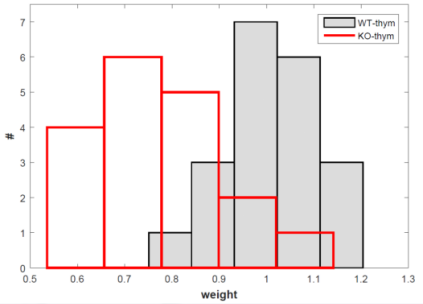
example: $\mathcal{N}(y_k, \sigma_k)$

now: comparing odds ratios!

$$\rho = \frac{P(\mathrm{M_A}|D,I)}{P(\mathrm{M_B}|D,I)} = \frac{P(D_m, D_n|\, A, I)}{P(D_m, D_n|\, B, I)} \frac{P(A|I)}{P(B|I)}$$

$$= \frac{(\mu_{max} - \mu_{min})\, \sigma_{max}}{\sqrt{2\pi}} \cdot \frac{N_m\, N_n}{N_{tot}} \cdot \frac{(\hat{\sigma})^{2-\boldsymbol{N_{tot}}}}{(\hat{\sigma}_n)^{2-\boldsymbol{N_n}}\, (\hat{\sigma}_m)^{2-\boldsymbol{N_m}}} \cdot \frac{P(A|I)}{P(B|I)}$$

$$\rho = \left[\frac{(\mu_{max} - \mu_{min})\, \sigma_{max}}{\sqrt{2\pi}}\right]^{K-1} \cdot \frac{(\hat{\sigma})^{2-N_{tot}}}{N_{tot}} \cdot \prod_{k=1}^{K} \frac{N_k}{(\hat{\sigma}_k)^{2-N_k}} \cdot \frac{P(A|I)}{P(B|I)}$$



$\boldsymbol{D_m}$: $\boldsymbol{N_m}$ **data points**
$\boldsymbol{D_n}$: $\boldsymbol{N_n}$ **data points**

$\boldsymbol{N_{tot} = N_n + N_m}$

note: - can be extended to any alternative distribution for $\mathrm{M_B}$

- alternative to a **two-tailed t-test** (if model is $\mathcal{N}(y_k, \sigma_k)$)

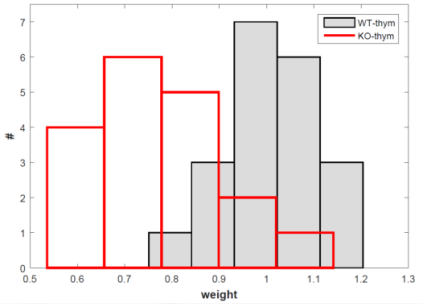
Bayesian Model Selection/Testing:

idea
curve fitting revisited
comparing distributions

example: $\mathcal{N}(y_k, \sigma_k)$

now: alternative to a **one-tailed t-test** (if model is $\mathcal{N}(y_k, \sigma_k)$)

Given D_1 and D_2, what is the probability that $\mu_1 > \mu_2$?



D_m: N_m data points
D_n: N_n data points

$N_{tot} = N_n + N_m$

μ_1 can have any value, but once set, it is a constrain for μ_2

$$P(\mu_1 > \mu_2 | D_1, D_2, I) = \int_{0\ or -\infty}^{\infty} \int_{0}^{\mu_1} P(\mu_1, \mu_2 | D_1, D_2, I)\, d\mu_2\, d\mu_1$$

$$= \int_{0\ or -\infty}^{\infty} \int_{0}^{\mu_1} P(\mu_1 | D_1, I)\, P(\mu_2 | D_2, I) d\mu_2\, d\mu_1$$

Bayesian Model Selection/Testing:

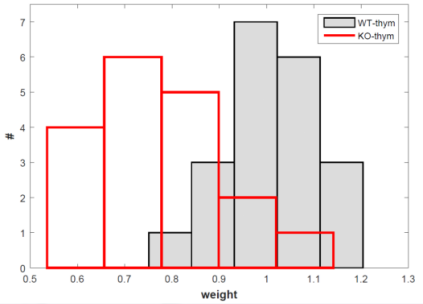
idea
curve fitting revisited
comparing distributions

example: $\mathcal{N}(y_k, \sigma_k)$

Given D_1 and D_2, what is the probability that $\mu_1 > \mu_2$?

$$P(\mu_1 > \mu_2 | D_1, D_2, I) = \int_{0\ or -\infty}^{\infty} \int_0^{\mu_1} P(\mu_1, \mu_2 | D_1, D_2, I)\, d\mu_1\, d\mu_2$$

$$= \int_{0\ or -\infty}^{\infty} \int_0^{\mu_1} P(\mu_1 | D_1, I)\, P(\mu_2 | D_2, I) d\mu_1\, d\mu_2$$



D_m:	N_m data points
D_n:	N_n data points

$N_{tot} = N_n + N_m$

we only compare means: marginalization over σ_k

$$P(\mu_k | D_k, I) = \int_0^{\infty} P(\mu_k, \sigma_k | D_k, I)\, d\sigma_k$$

$$\sim \int_0^{\infty} P(D_k | \mu_k, \sigma_k, I) P(\mu_k, \sigma_k | I)\, d\sigma_k$$

max ent:

$$P(\mu_k, \sigma_k | D_k, I) = \frac{P(D_k | \mu_k, \sigma_k, I)\, P(\mu_k, \sigma_k | I)}{P(D_k, I)}$$

Bayes' theorem

Bayesian Model Selection/Testing:

idea
curve fitting revisited
comparing distributions

example: $\mathcal{N}(y_k, \sigma_k)$

Given D_1 and D_2, what is the probability that $\mu_1 > \mu_2$?

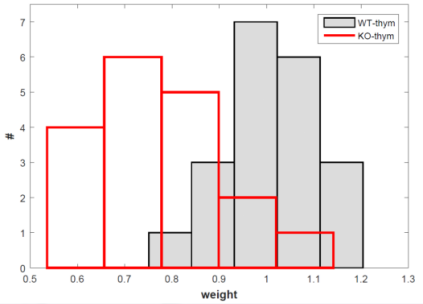
$$P(\mu_1 > \mu_2 | D_1, D_2, I) = \int_{0\ or -\infty}^{\infty} \int_0^{\mu_1} P(\mu_1, \mu_2 | D_1, D_2, I)\, d\mu_1\, d\mu_2$$

$$= \int_{0\ or -\infty}^{\infty} \int_0^{\mu_1} P(\mu_1 | D_1, I)\, P(\mu_2 | D_2, I) d\mu_1\, d\mu_2$$

$$P(\mu_k | D_k, I) \sim \int_0^{\infty} P(D_k |\, \mu_k, \sigma_k, I) P(\mu_k, \sigma_k | I)\, d\sigma_k$$

likelihood function
→ the actual model

$$P(D_k |\, \mu_k, \sigma_k, I) = \frac{1}{(2\pi {\sigma_k}^2)^{N_k/2}} \cdot e^{-\frac{1}{2{\sigma_k}^2} \sum_{i=1}^{N_k} (x_i - \mu_k)^2}$$



D_m: N_m data points
D_n: N_n data points

$N_{tot} = N_n + N_m$

Bayesian Model Selection/Testing:

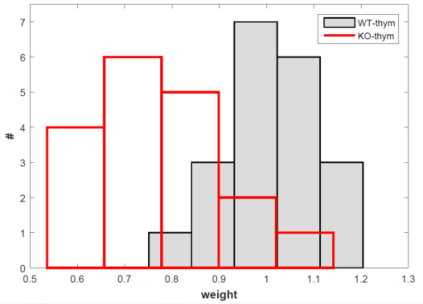
idea
curve fitting revisited
comparing distributions

example: $\mathcal{N}(y_k, \sigma_k)$

Given D_1 and D_2, what is the probability that $\mu_1 > \mu_2$?

again, approximating the integral for large N

$$P(\mu_1 > \mu_2 | D_1, D_2, I) \approx \frac{1}{S_{tot}\sqrt{2\pi}} \int_0^\infty \exp\left[-\frac{1}{2\,S_{tot}^2}(\zeta - \hat{z})^2\right] d\zeta$$



D_m: N_m data points
D_n: N_n data points

$N_{tot} = N_n + N_m$

where:

$$S_{tot}^2 = \frac{S_1^2}{N_1} + \frac{S_2^2}{N_2}$$

$$\hat{z} = \hat{\mu}_1 - \hat{\mu}_2$$

$$\zeta = \mu_1 - \mu_2$$

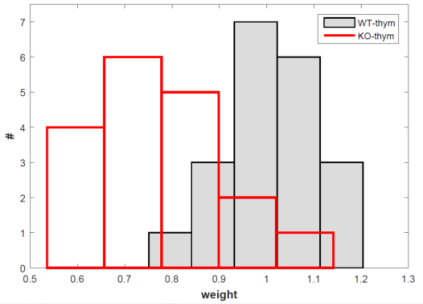
$$\hat{\mu}_k = \frac{1}{N_k}\sum_{i=1}^{N_k} x_{i,k} \qquad S_k^2 = \frac{1}{N_k - 1}\sum_{i=1}^{N_k}\left(x_{i,k} - \hat{\mu}_k\right)^2$$

Bayesian Model Selection/Testing:

idea
curve fitting revisited
comparing distributions

example: $\mathcal{N}(y_k, \sigma_k)$

Given D_1 and D_2, what is the probability that $\mu_1 > \mu_2$?



again, approximating the integral for large N

$$\boxed{P(\mu_1 > \mu_2 | D_1, D_2, I) \approx \frac{1}{S_{tot}\sqrt{2\pi}} \int_0^\infty \exp\left[-\frac{1}{2\,S_{tot}^2}(\zeta - \hat{z})^2\right] d\zeta}$$

D_m: N_m data points
D_n: N_n data points

$N_{tot} = N_n + N_m$

where:

$$S_{tot}^2 = \frac{S_1^2}{N_1} + \frac{S_2^2}{N_2} \qquad \hat{\mu}_k = \frac{1}{N_k}\sum_{i=1}^{N_k} x_{i,k}$$

$$\hat{z} = \hat{\mu}_1 - \hat{\mu}_2$$

$$\zeta = \mu_1 - \mu_2 \qquad S_k^2 = \frac{1}{N_k - 1}\sum_{i=1}^{N_k}\left(x_{i,k} - \hat{\mu}_k\right)^2$$

note: - more detailed derivation: Sivia, “Data Analysis”, page 53

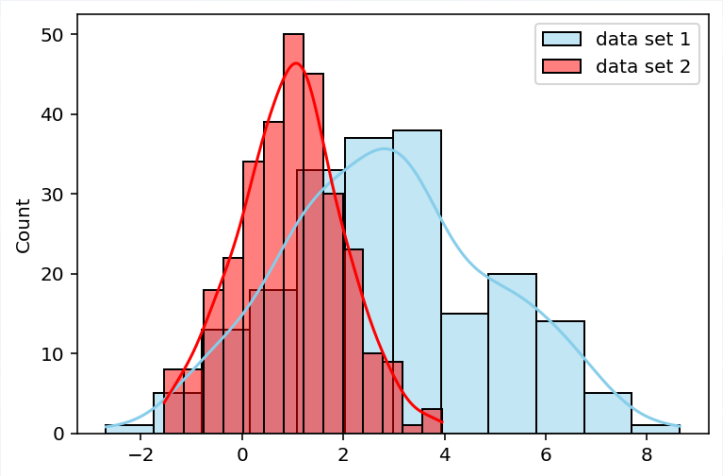
- changing μ_1 with μ_2 changes $P(\mu_1 > \mu_2 | D_1, D_2, I)$ to $1 - P(\mu_1 > \mu_2 | D_1, D_2, I)$

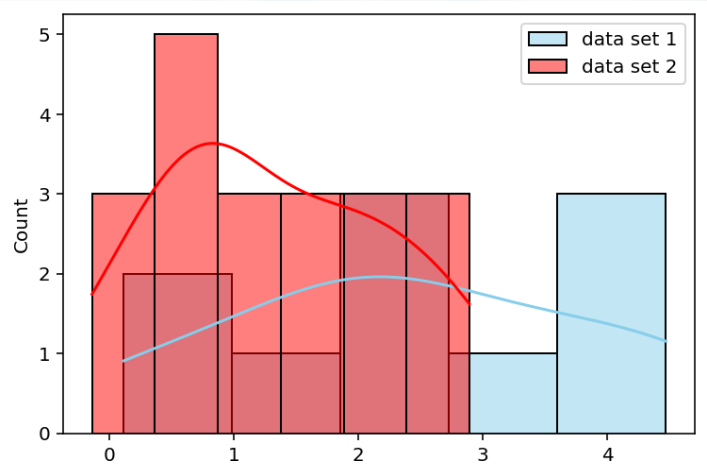
Bayesian Model Selection/Testing:

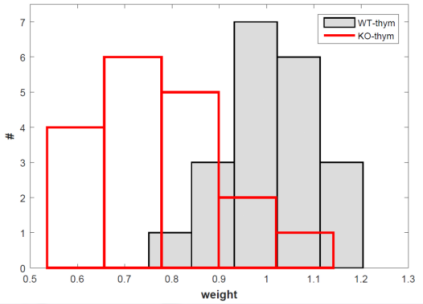
idea
curve fitting revisited
comparing distributions

example: $\mathcal{N}(y_k, \sigma_k)$

Check out: `mean_var_analysis_example.py`







D_m: N_m data points
D_n: N_n data points

$N_{tot} = N_n + N_m$

```
M.ModelSelection()
```

```
M.T_Test()
(2.144, 0.060)
```

p – value

t – value: $t = \frac{\hat{\mu}_1 - \hat{\mu}_2}{\sqrt{\frac{\hat{\sigma}_1^2}{N_1} + \frac{\hat{\sigma}_2^2}{N_2}}}$

```
M.ModelSelection()
0.711
```

$$\rho = \frac{P(\mathrm{M_A} = from\ one\ pdf | D, I)}{P(\mathrm{M_B} = from\ two\ pdfs | D, I)}$$

```
M.P_Means()
(0.99, 0.01)
```

$P(\mu_1 > \mu_2 | D_1, D_2, I)$ and $P(\mu_1 < \mu_2 | D_1, D_2, I)$

Bayesian Model Selection/Testing

Thank you very much for your attention!